Are Time-Series Foundation Models Operationally Reliable? Conformal Calibration and Decision Value for History-Only Day-Ahead Load, Solar and Wind Forecasting

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***Abstract*—Grid operators size reserves and commit storage from forecast quantiles, yet time-series foundation models (TSFMs) are judged mainly on point accuracy. We ask whether their uncertainty can be used as delivered. Four TSFMs (Chronos-2, TimesFM-2.5, Moirai-2.0, TiRex) and four trained baselines forecast day-ahead load, solar and wind for ERCOT and CAISO from EIA-930 history only, on a ten-month test period that starts after every checkpoint was published. Each model's quantiles go through four conformal calibration methods and then through a newsvendor reserve rule and a battery-firming program. First, the leading TSFMs have the lowest point error on load and wind, but their margin over the best trained baseline is small on wind and not always significant. Second, native 80% bands cover 78% on average, yet only 15% of model-series cases pass a coverage test, and the worst hour of the day misses by 14.7 points on average. Third, a per-hour conformal layer fixes average coverage at little extra width and lowers reserve cost for all 8 models. Reserve cost is a quantile loss at the operating level plus a bias term, and its ranking departs from the point-accuracy ranking on some series.**

***Index Terms*—Foundation models, probabilistic forecasting, conformal prediction, operating reserves, energy storage.**

I. Introduction

Zero-shot time-series foundation models now match or beat task-specific models on many energy benchmarks [1]–[3]. Most of those benchmarks report median errors and at most a proper score such as CRPS, but operators use the tails: reserve requirements, battery commitments and risk limits are set from quantiles. An accurate but overconfident model under-procures reserve, and an underconfident one wastes capacity.

Three earlier studies are closest to ours. Achour et al. applied split conformal prediction to earlier-generation TSFMs (Lag-Llama, Chronos, TimesFM) on general benchmarks and ERCOT load [4]. Beichter et al. fine-tuned one TSFM for one feeder decision [5]. Simeone measured interval coverage of Chronos-2 and Moirai-2 on ERCOT load and set reserve margins from Chronos-Bolt quantiles [3]; he found Chronos-2 slightly conservative at a nominal 90% level and Moirai-2 overconfident. To our knowledge, nobody has yet measured calibration and operating cost for four current TSFMs across load, solar and wind, under several conformal methods, with hour-of-day coverage and decision cost reported for every model. Our native-calibration result need not conflict with Simeone's: he evaluates 90% intervals with contexts up to 2048 h on ERCOT load in 2020–2024, we the 80% band with a 28-day context after checkpoint release; Chronos-2's average coverage here is also near nominal (0.78), and our tests act on single series and hours.

We ask: **RQ1** how accurate are current zero-shot TSFMs against trained models on history-only day-ahead load, solar and wind? **RQ2** are their native intervals calibrated, and which conformal method restores nominal coverage with the narrowest intervals as conditions change between winter and summer? **RQ3** when intervals drive reserve sizing and battery scheduling, which model and calibration pair is cheapest, and does that follow the accuracy ranking? Contributions:

1) A leakage-controlled benchmark of four TSFMs, four baselines and the balancing authority's load forecast on six EIA-930 series, tested after every checkpoint's publication.
2) A comparison of native and four conformal calibrations at equal nominal levels, including coverage by hour of day and miss clustering.
3) A decision-value evaluation of reserve cost and battery profit with paired block-bootstrap confidence intervals, and a decomposition that ties reserve cost to the quantile loss at the operating level.
4) Code, pinned configurations and outputs that rebuild every table and figure. https://github.com/TanishqX-404/tsfm-grid-calibration

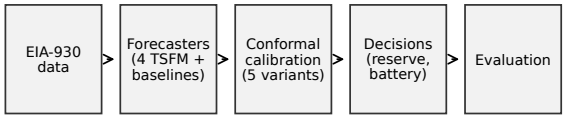


Fig. 1. Pipeline. Every model writes the same quantile table; calibration and decision code is shared, so cost differences come only from the forecasts.

II. Data and Protocol

Data. We use hourly EIA-930 demand and solar and wind generation [6] for ERCOT and CAISO from the PUDL tables [7] (six series). Timestamps are in local standard time, and gaps of up to 3 h are interpolated. Solar and wind are normalized by a trailing 90-day maximum (a causal capacity proxy) and load by its training mean. Training runs from July 2019 to December 2024, validation from January to October 2025 and test from 1 November 2025 to 31 August 2026 (304 daily origins; September and October are not in the test year). Days with longer gaps are dropped for every model. EIA-930 lacks 5 December 2025 for ERCOT, which removes 30 ERCOT test days.

Information set. Forecasts are *history-only*, with no weather or other covariates. The forecast for day D is issued at midnight using the 28 days before D and covers the 24 hours of D (lead times 1–24 h, so horizon equals hour of day). Market day-ahead forecasts are made about 14 h earlier, when the ERCOT and CAISO day-ahead markets close on the morning of $D-1$. Our early horizons are therefore close to nowcasts, which flatters every model, and persistence-like ones most. Solar night hours, defined causally from the data (CAISO's EIA-930 solar profile drifts by about 1.5 h over 2022–2025), are excluded from accuracy, coverage and width. In the reserve problem they carry zero reserve and zero error.

Models. The TSFMs are Chronos-2 [8], TimesFM-2.5 [9], Moirai-2.0-R-small [10] and TiRex [11], run zero-shot from pinned checkpoints. The baselines are each trained on their own series only: seasonal naive (lag 24 h for solar and wind, 168 h for load, with empirical residual quantiles), LightGBM quantile regression [12] on 24/48/168-h lags and calendar features (Optuna, 30 trials), and N-HiTS [13] and PatchTST [14] with a multi-quantile loss (three seeds each). N-HiTS memorized its single training series and its quantiles collapsed on validation, so for load and wind it uses early stopping on the last 90 training days. All models output quantiles $0.1, \ldots, 0.9$, and the median is the point forecast. This common grid is set by TimesFM-2.5 and TiRex, whose heads return only these levels; Chronos-2, which also predicts 0.05 and 0.95, was queried at the same levels. We also report the balancing authority's day-ahead load forecast.

Leakage control. Table I lists when each checkpoint's weights were committed; all precede the test start. No documented pretraining corpus contains EIA-930 or CAISO data. An older ERCOT load set, a held-out Chronos benchmark [15], may have been used by Chronos-2 and TiRex but predates our test period.

TABLE I
Checkpoint dates (weights file at the pinned revision).

Model	Weights committed	Report (arXiv)
TiRex	2025-05-26	2505.23719
Moirai-2.0	2025-08-06	2511.11698
TimesFM-2.5	2025-10-01	2310.10688
Chronos-2	2025-10-30	2510.15821

Protocol. All settings were chosen on validation, frozen and tagged: the LightGBM hyper-parameters, the calibration step sizes, the fixed-reserve share, and each model's calibration variant and battery interval level in Table IV. The test period was evaluated once. The TSFM test forecasts reproduced bit-for-bit in a second run.

III. Calibration and Decision Layers

Calibration. Conformalized quantile regression (CQR) [16] scores each past day by how far the outcome fell outside the model's interval, $s = \max(\hat{q}^{lo} - y,\ y - \hat{q}^{hi})$, and widens (or narrows) the interval by a quantile Q of past scores: $[\hat{q}^{lo} - Q,\ \hat{q}^{hi} + Q]$. Each calibrator is fit separately per series and per hour of day and updated daily from days whose outcomes are known. The four variants differ only in how Q is chosen. *Split CQR* takes the empirical quantile of the last $W = 90$ days. *NexCP* [17] weights older days less (0.99^{age}). *ACI* [18] raises the target miscoverage after each covered day and lowers it after each miss, $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$. *QT* is the proportional, quantile-tracking term of conformal PID control [19]: it moves Q itself by a fixed step after each hit or miss. We use neither the integral nor the forecasting term, so QT is not the full PID controller. Energy series are not exchangeable, so the coverage of split CQR and NexCP is empirical. ACI and quantile tracking guarantee long-run average coverage for any sequence (quantile tracking for bounded scores). The native band $[\hat{q}_{0.1}, \hat{q}_{0.9}]$ is the only native interval the models provide, so we compare native and conformal intervals at the 80% level and report conformal intervals alone at 90%. One-sided bounds for the decision layer apply the same method to the nearest native quantile (0.9 or 0.1), so native reserve bounds sit at 0.9 even when the decision needs 0.95.

Tests. The Kupiec test checks whether the share of misses equals the nominal rate [20], and the Christoffersen test checks whether a miss makes the next miss more likely [21]. Kupiec, applied to all hourly misses of a series, assumes independent misses, so its rejection rates are inflated; we therefore add block-bootstrap CIs. Christoffersen is applied to each hour's daily miss sequence (median p over hours).

Reserve sizing. With reserve cost c_R and shortfall penalty c_S, the expected-cost-optimal upward reserve is the $\tau^* = 1 - c_R/c_S$ quantile of the forecast error (newsvendor). Reserve R_h is the calibrated one-sided bound at τ^* minus the median (upper bound for load, lower for solar and wind). Daily cost

is $J = \sum_h [c_R R_h + c_S(e_h - R_h)^+]$, with error e_h in the reserve direction, in normalized units. Whenever the bound lies beyond the median,

$$J/c_S = \sum_h \rho_{\tau^*}(e_h - R_h) + (1-\tau^*)\sum_h e_h, \qquad (1)$$

where ρ_τ is the pinball loss. Reserve cost is therefore the quantile loss of the τ^* bound plus a bias term. We report $\tau^* = 0.95$ ($c_S/c_R = 20$) and, as a simple reference, a fixed rule $R_h = k\hat{y}_h$ with k tuned on validation.

Battery firming. A solar or wind plant of capacity 1 with a 0.5/1.0 battery (90% round-trip) commits a day-ahead schedule by a linear program on the calibrated lower bound under a stated two-level time-of-use price. A greedy real-time rule covers deviations; shortfalls pay three times the price and surplus sells at half. Profit is reported as a share of an oracle that plans on the true outcome.

IV. Results

RQ1: accuracy. Table II gives test nMAE. Chronos-2, Moirai-2.0 and TiRex have the lowest nMAE on load and wind, but the margin over the best trained baseline is small, especially on wind. On ERCOT wind Chronos-2 reaches 0.121 against 0.172 for LightGBM and 0.204 for seasonal naive, which is 30% lower than LightGBM (17% on CAISO wind), though most of this margin comes from the first hours (Section V). The fairer comparison is PatchTST, the strongest trained model on both wind series: Chronos-2 is 7% lower on ERCOT wind (0.121 vs. 0.130) and 2% lower on CAISO wind (0.111 vs. 0.113). The pinball-loss tests in Table II (Diebold–Mariano with Harvey correction [22], [23], Holm-adjusted, 5%) are made against the best model in each column. By that test the best trained baseline is significantly worse on ERCOT load, CAISO load and ERCOT wind, and statistically tied with the best model on CAISO wind. Chronos-2 beats LightGBM significantly on 4 of six series; only 2 of 18 comparisons among the three leading TSFMs are significant. On solar, lagged LightGBM is as accurate as the TSFMs (CAISO: 0.067 vs. 0.070). ERCOT's own day-ahead load forecast, which uses weather, remains the most accurate (0.024 vs. 0.035). The CAISO operator row is not comparable, because EIA-930 CAISO demand has diverged from CAISO's forecast definition since 2022.

RQ2: calibration. Table III pools eight models and six series. Averages hide the problem: the native 80% band covers 0.784 on average, and PatchTST and TiRex are close to nominal. But 71% of model-series pairs under-cover, only 15% pass the Kupiec test, and the worst hour of the day misses by 14.7 points on average (up to 43). At the same 80% level, conformal intervals cover 0.801–0.807 and are only 6% wider. They score slightly better on the interval (Winkler) score (0.397–0.404 vs. 0.408) and cut the worst-hour gap to 8.6–9.3 points. At 90% they cover 0.900–0.906 on average (0.87–0.93 over series). The four methods differ little (width within 2%); the adaptive ones mainly reduce miss clustering. We recommend the simplest, split CQR with a 30–90-day window, since a shorter window gives better hour-of-day coverage (worst-hour gap 4.2, 5.9 and 7.4 points for $W = 30, 90, 180$ days at 90%). With moving-block bootstrap 95% CIs (blocks of 3–28 days), the pooled native 80% CI excludes 0.80 at every block length (0.770–0.792 to 0.763–0.799), while every pooled conformal CI contains its nominal level. Per pair, the native CI contains 0.80 in 54–67% of pairs (94–100% for conformal), so Kupiec overstates native failure. ACI capped its interval on 0.9% of test days, and removing EIA-imputed hours changes coverage by at most 0.3 points.

RQ3: decision value. Table IV reports reserve cost at $\tau^* = 0.95$ per series. The mean columns average over the 274 days on which all six series have forecasts, the same days the paired bootstrap uses. Calibration lowers reserve cost for all 8 models, by 0.19–1.48 normalized units per day (paired moving-block bootstrap [24], 7-day blocks, 95% intervals excluding zero). Most of this gain comes from reaching the 0.95 level: at $\tau^* = 0.90$, where native $\hat{q}_{0.9}$ exists, calibration lowers cost by only 1.7% on average (0–7.9% per model), against 11.2% at 0.95. Calibrated intervals are 27–38% cheaper than the fixed-percentage rule. Chronos-2 has the lowest mean cost (4.03). TiRex costs 0.12 more per day (95% CI −0.01 to 0.26, not significant) and Moirai-2.0 costs

TABLE II

Test nMAE (normalized MAE of the median; solar daytime only) and mean native 80% coverage. Bold: lowest per series (ties at three decimals are both bold); underlined: pinball loss not significantly worse than the best (DM, Holm, 5%). The operator forecast is a point forecast with no native interval.

Model	ERCOT L	ERCOT S	ERCOT W	CAISO L	CAISO S	CAISO W	Cov_{80}
Seasonal naive	0.095	0.104	0.204	0.066	0.075	0.155	0.78
LightGBM	0.051	0.097	0.172	0.034	**0.067**	0.134	0.79
N-HiTS	0.040	0.126	0.135	0.042	0.085	0.114	0.73
PatchTST	0.038	0.116	0.130	0.035	0.095	0.113	0.80
Chronos-2	0.036	**0.095**	**0.121**	0.032	0.070	0.111	0.78
Moirai-2.0	0.036	**0.095**	0.124	**0.031**	0.070	0.110	0.79
TimesFM-2.5	0.044	0.103	0.127	0.037	0.076	0.112	0.78
TiRex	**0.035**	0.100	**0.121**	0.033	0.074	**0.109**	0.81
Operator (BA)	0.024	–	–	0.096	–	–	–

TABLE III

Calibration pooled over eight models and six series (test). IS: interval (Winkler) score; Worst hr: mean largest hour-of-day coverage gap from nominal; Kup./Chr.: share of model-series pairs passing the Kupiec and Christoffersen tests at 5%. QT: quantile tracking (proportional term of conformal PID control only).

Variant	Cov.	Range	Width	IS	Worst hr (pp)	Kup. (%)	Chr. (%)
Target 80% (native band $[\hat{q}_{0.1}, \hat{q}_{0.9}]$ available)							
Native	0.784	0.66–0.87	0.276	0.408	14.7	15	71
Split CQR	0.801	0.75–0.83	0.293	0.404	8.6	69	71
NexCP	0.807	0.78–0.83	0.293	0.402	9.3	44	75
ACI	0.804	0.77–0.84	0.294	0.400	9.1	62	83
QT	0.802	0.78–0.83	0.290	0.397	8.6	71	90
Target 90% (no native interval)							
Split CQR	0.900	0.87–0.91	0.375	0.482	5.9	71	81
NexCP	0.906	0.89–0.92	0.378	0.480	6.1	56	85
ACI	0.905	0.88–0.93	0.379	0.478	5.9	65	90
QT	0.900	0.88–0.92	0.370	0.475	7.3	60	90

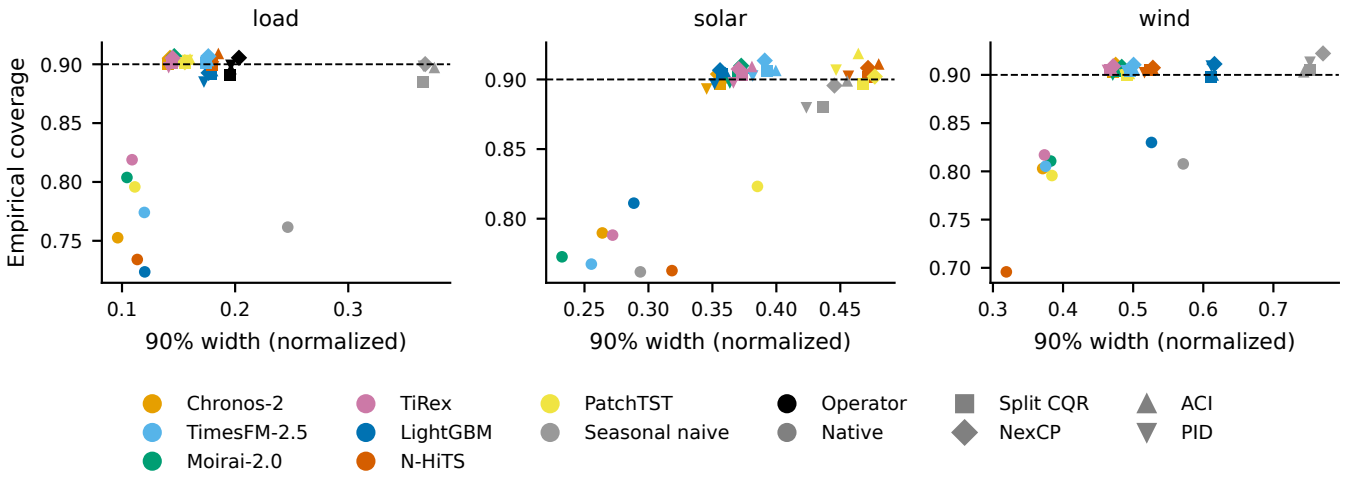


Fig. 2. Coverage of 90% intervals against normalized width (test), averaged over the two regions; the native points are the 80% band. Colour: model; marker: calibration variant; dashed: nominal.

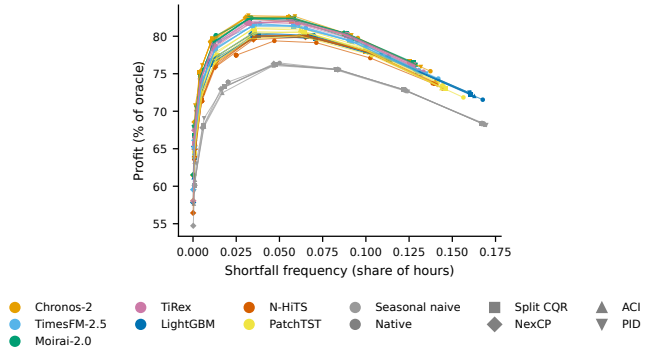


Fig. 3. Battery firming frontier (test, mean over solar and wind series): profit as % of oracle against shortfall frequency; each line traces one model and calibration pair across interval levels 0.5–0.99.

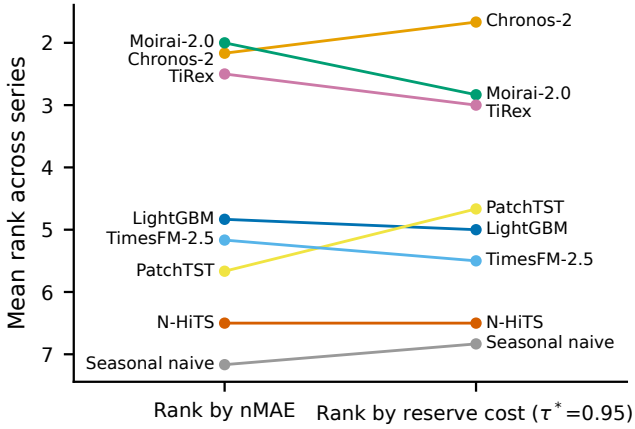


Fig. 4. Mean rank across series by nMAE (left) and by reserve cost at $\tau^* = 0.95$ with the validation-selected calibration (right), test period.

0.18 more (CI 0.08–0.28); 6 of 7 differences are significant. Shortfall occurs in 5.1% of hours on average, close to the 5% implied by τ^*. For ERCOT load the operator's point forecast with conformal intervals is cheaper than any model (regret 1.11 vs. 1.99); without calibration it cannot size reserve at all (regret 23.2). In battery firming (Fig. 3) the TSFMs recover 81.9–82.7% of oracle profit (best: Moirai-2.0), against 80.4% for LightGBM and 76.7% for seasonal naive. The calibration variant changes profit by at most 1.8 points, much less than the commitment level does.

Accuracy vs. cost. Equation (1) makes the link between cost and quantile loss close to an identity: reserve cost is the pinball loss of the τ^* bound plus a bias term, so a model's cost rank mostly follows its quantile-loss rank at τ^*, and only indirectly its median error. Across series the Spearman correlation between cost rank and nMAE rank is 0.45–1.00, and between cost rank and the pinball loss of the τ^* bound it is 0.69–1.00 (Fig. 4). The lowest-nMAE model is not the cheapest on 3 of six series (4 if Moirai-2.0 wins the ERCOT solar tie), but the cost gap is significant only under that tie-break. With eight models the rank correlations are imprecise, so we claim only that nMAE is an unreliable proxy for operating cost.

Ablations. A 7-day TSFM context instead of 28 days raises nMAE from 0.077 to 0.081 and reserve cost from 4.13 to 4.48; a 90-day context gives 0.076 and 7% narrower calibrated intervals at equal cost. Baseline seed spread is small (regret s.d. ≤ 0.28).

V. Discussion and Limitations

From history alone, zero-shot TSFMs have the lowest point error on load and wind among the models we tested, but their quantiles should not be used as reserve requirements as delivered. A per-hour conformal layer that runs in seconds on a CPU fixes average coverage and lowers reserve cost for every model, most for the worst-calibrated ones. **Protocol**

TABLE IV

TEST RESERVE COST AT $\tau^* = 0.95$ (NORMALIZED UNITS PER DAY). PER SERIES: VALIDATION-SELECTED CALIBRATION, ALL DAYS OF THE SERIES. MEAN: THE 274 COMMON DAYS (NOT THE AVERAGE OF THE PER-SERIES COLUMNS), WITH CALIBRATED, NATIVE AND FIXED-RULE RESERVE. BATTERY: PROFIT AS % OF ORACLE (SOLAR AND WIND). BOLD: BEST; UNDERLINED: NOT SIGNIFICANTLY DIFFERENT FROM THE BEST (PAIRED BLOCK BOOTSTRAP, 95%).

	Reserve cost, validation-selected calibration						Mean (common days)			Battery
Model	ERCOT L	ERCOT S	ERCOT W	CAISO L	CAISO S	CAISO W	Cal.	Native	Fixed	(% orc.)
Chronos-2	**2.50**	**3.05**	7.27	2.16	**3.22**	5.69	**4.03**	4.28	5.85	82.4
TiRex	2.54	3.58	7.12	2.28	3.70	**5.61**	4.15	4.34	5.88	82.5
Moirai-2.0	2.82	3.43	**7.09**	**2.08**	3.71	5.80	4.21	4.51	5.76	**82.7**
PatchTST	2.75	3.79	7.44	2.53	3.69	6.27	4.47	4.89	6.55	81.5
TimesFM-2.5	3.07	3.49	7.55	2.54	3.96	5.99	4.49	5.02	6.34	81.9
N-HiTS	2.96	3.88	8.07	2.98	3.81	6.28	4.73	6.20	6.47	80.1
LightGBM	3.69	3.47	8.57	2.44	3.43	6.90	4.84	5.46	7.85	80.4
Seasonal naive	6.93	3.58	10.65	4.67	3.46	9.56	6.58	7.82	9.55	76.7

bias. LightGBM and seasonal naive use lags $\geq$24 h, while the neural models see the context up to midnight, which favours them at short lead times. Over the six series the best TSFM's nMAE is 57% below LightGBM's for hours 1–6 but 4.3% below for hours 13–24, the lead times nearest a market forecast (ERCOT wind: 7.9%); against the best trained model, 12.7% and 1.7%. A TSFM still leads at hours 13–24 on five of six series. The RQ1 margins are thus upper bounds for a market-close issue; calibration, fit per hour, is unaffected. Limitations: TSFMs are used zero-shot; fine-tuning (supported by Chronos-2) may change accuracy and calibration. Forecasts are history-only and issued at midnight rather than at market close, so they understate the lead time and omit the weather forecasts operators rely on, especially for solar and wind. The fixed-percentage rule is a weak reference, and empirical-residual and operator-style dynamic reserve baselines remain to be added. Prices are stated, not market prices; results cover two balancing authorities and ten months; baselines are single-series models. The CAISO demand definition in EIA-930 limits the operator comparison to ERCOT.

VI. CONCLUSION

Current foundation models have the lowest history-only point error on load and wind, with a small margin on wind that shrinks at longer lead times, but their quantiles are rarely calibrated as delivered. Conformal calibration is cheap, restores nominal coverage with little extra width and lowers reserve cost for every model; Chronos-2 and TiRex were cheapest. As Eq. (1) implies, cost ranks follow the quantile loss at the operating level, not nMAE.

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